

Introduction to Scientific Computing

ENGR1050 – Lecture 5

Fall 2026

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University of Pennsylvania



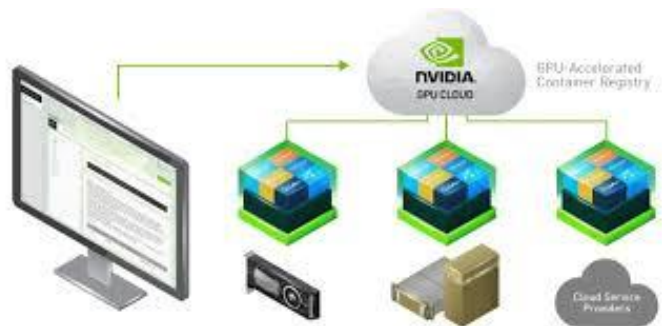
Introduction to Microcontrollers

To drive a circuit, we can't use Colab – it lives in the cloud and can't touch a circuit in real life!

Today we will install Thonny, which is a tool to run Python on your laptop that we can use to drive circuits and make robots

Today you only have two jobs: get Thonny installed and working on your computer, and use it to blink an LED light

Three places so far for you to run code – understand what they mean



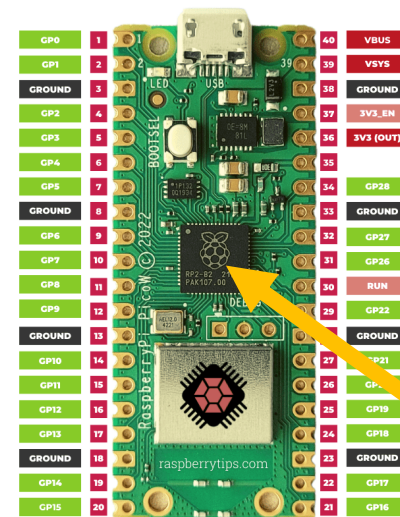
Runs in the cloud

You connect through the internet to see program output, but its running on some GPU in Google's basement



Run on your laptop

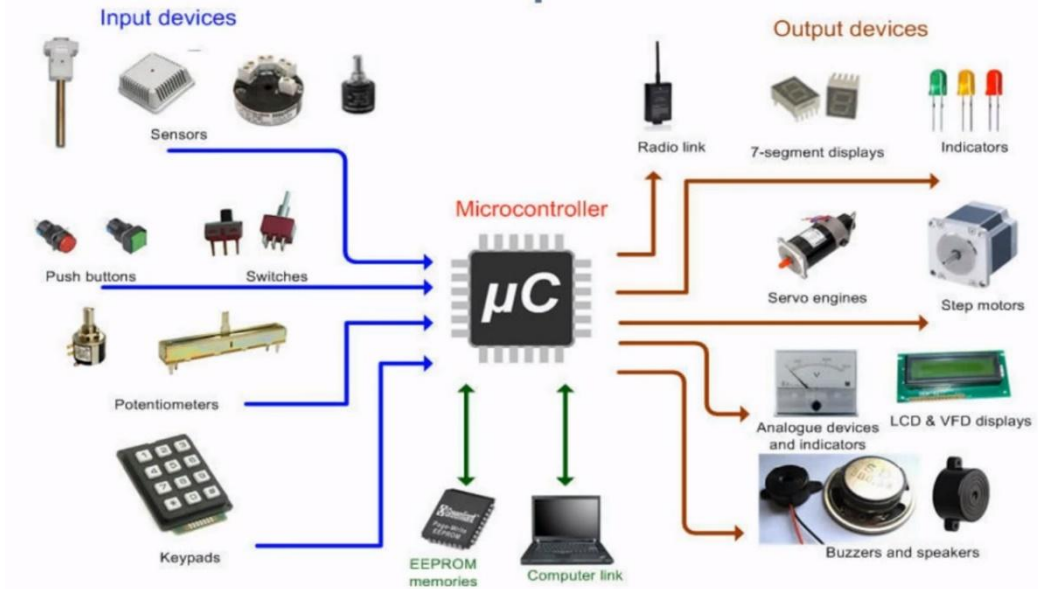
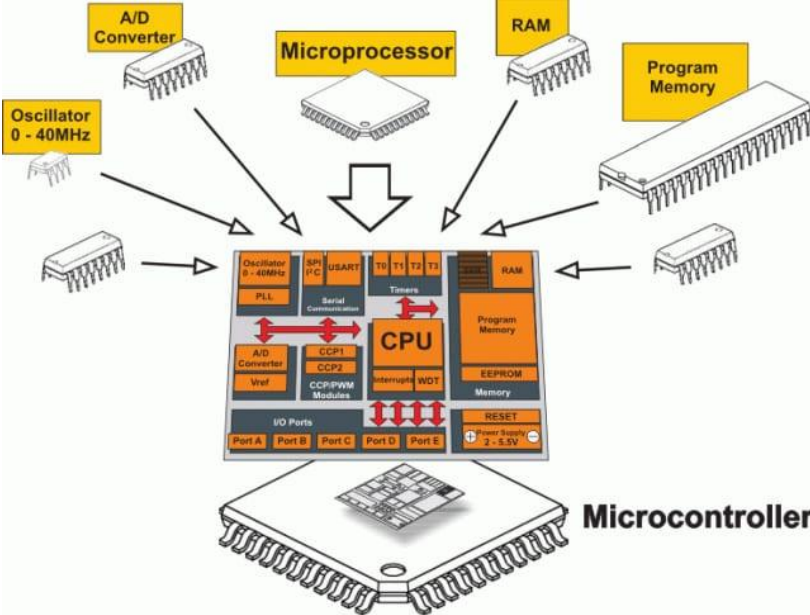
Today we'll use Thonny to run directly on your laptop



Run on a microcontroller

We'll show how to run a program on this little black chip
This is good for steering robots, drones, and other little mechatronic systems

What is a microcontroller?



Step 1

Install Thonny

Installation instructions

Sep 14

Lecture 5: Microcontrollers and the Thonny IDE

Slides (PDF)

Exercise submission

Thonny installation (Windows)

Thonny installation (Mac)

Pico tutorial code



Installing Thonny on Windows

Up to now, every line of Python you have written ran in Colab, which means it ran on a computer in a Google data center. To drive a circuit or a robot, the program has to run on *your* laptop, because your laptop is the thing with a USB cable plugged into the microcontroller. Thonny is a small, free program that runs Python on your own computer. It is the only tool you need for the hardware part of this course.

These directions assume you have never installed a program from the internet before. Nothing here is hard, but there are several steps where the computer will ask you a question, and the directions tell you what to answer. Follow them in order and do not skip any. The whole thing takes about ten minutes.

Using a Mac? Stop here and use the [Mac instructions](#) instead. The steps are different.

What you need before you start

- A Windows laptop (Windows 10 or Windows 11) that you can install programs on. If it is a school-managed laptop that blocks installs, come to office hours.
- An internet connection.
- About 200 MB of free disk space.

Installing Thonny on a Mac

Up to now, every line of Python you have written ran in Colab, which means it ran on a computer in a Google data center. To drive a circuit or a robot, the program has to run on *your* laptop, because your laptop is the thing with a USB cable plugged into the microcontroller. Thonny is a small, free program that runs Python on your own computer. It is the only tool you need for the hardware part of this course.

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Using Windows? Stop here and use the [Windows instructions](#) instead. The steps are different.

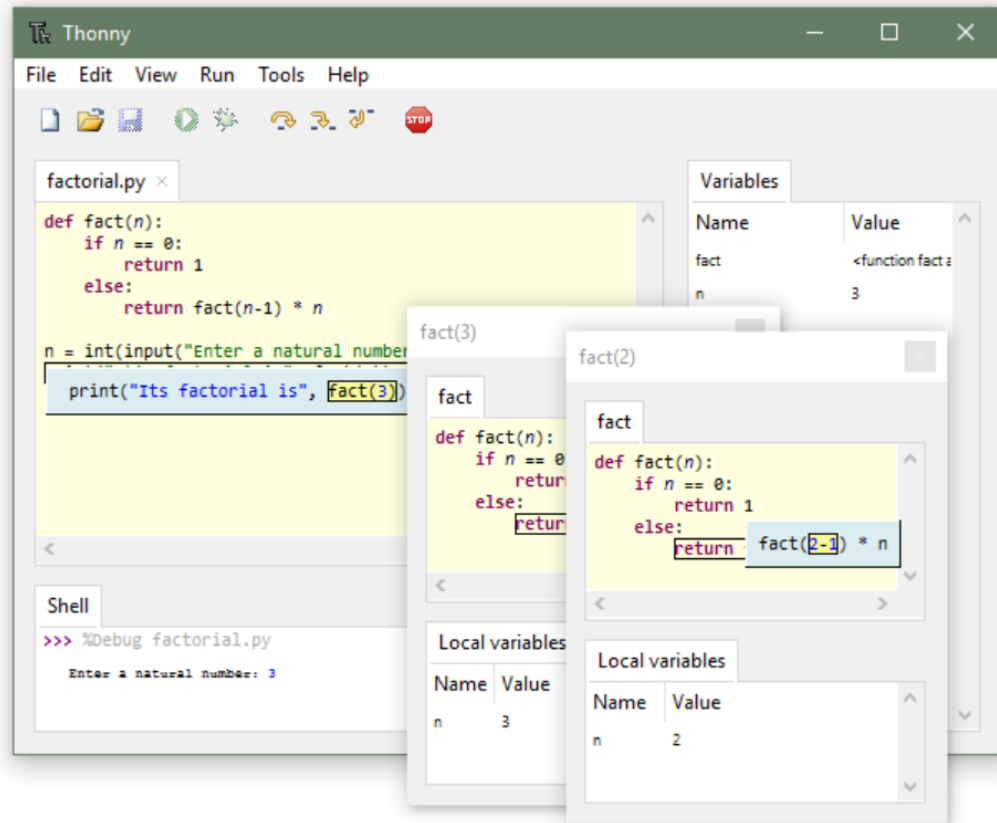
What you need before you start

- A Mac running macOS 10.15 (Catalina) or newer. Anything bought in the last several years qualifies.
- The password you use to log in to the Mac. The installer will ask for it.
- An internet connection.

Installation instructions

Thonny

Python IDE for beginners



Download version [5.0.0](#) for
[Windows](#) • [Mac](#) • [Linux](#)

Official downloads for Windows

Installer with Python 3.14 for Intel or AMD computers, requires Windows 10 or 11
[thonny-5.0.0-x64.exe \(24 MB\)](#) 👉 *recommended for you*

Installer with Python 3.14 for Arm (e.g. Snapdragon) computers
[thonny-5.0.0-arm64.exe \(23 MB\)](#)

Portable variant with Python 3.14 for Intel or AMD computers, also usable on Arm
[thonny-py38-5.0.0-windows-portable-x64.zip \(36 MB\)](#)

Re-using an existing Python installation (for advanced users)
pip install thonny

Official downloads for macOS

Installer with Python 3.14 for Intel Macs
[thonny-5.0.0-x64.pkg \(47 MB\)](#)

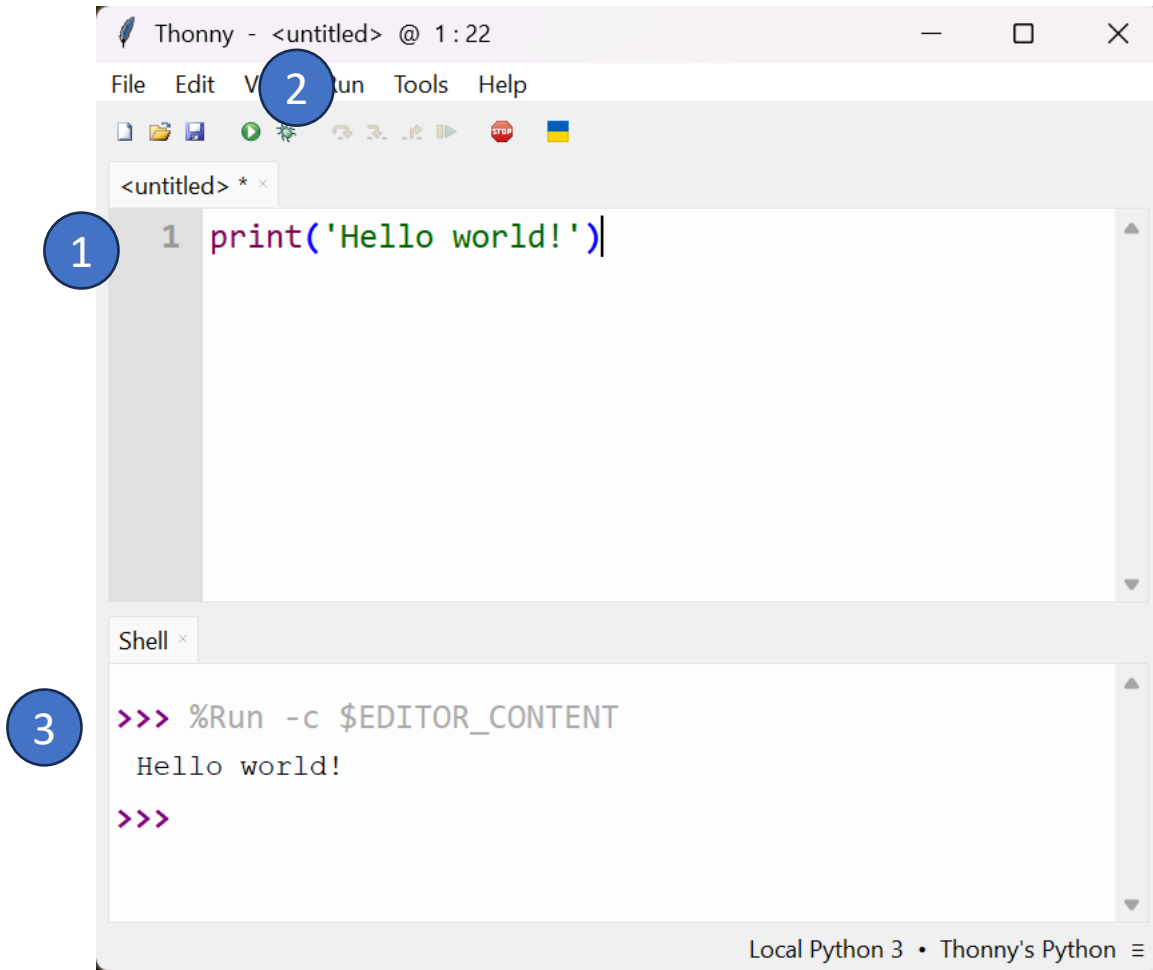
Installer with Python 3.14 for Apple Silicon Macs (M1, M2 ...)
[thonny-5.0.0-arm64.pkg \(47 MB\)](#)

Re-using an existing Python installation (for advanced users)
pip install thonny

3rd party distributions (may have older version)

Downloading and launching the installer with Homebrew
brew install thonny

Installation instructions



- 1 Type your python commands into the big white box. This is equivalent to a single code block of the Jupyter notebooks we've been using in colab. A .py file executes once sequentially, top to bottom
- 2 Click the green arrow to run the program
- 3 The output of the program will be displayed in the box at the bottom of the screen. Take a quick picture on your phone. One half of today's assignment is to upload that picture to Canvas so that we can confirm you have Thonny set up and ready to run the future labs

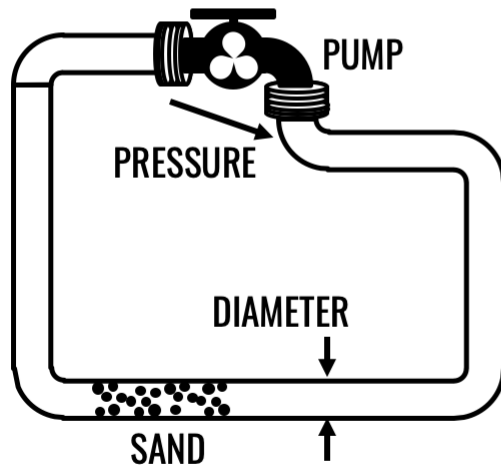
Step 2

Flash an LED

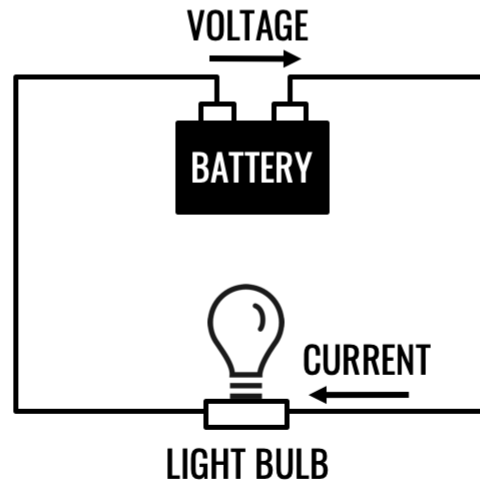
What is a circuit?

$$\text{Voltage} = \text{Current} \times \text{Resistance}$$
$$(V = I \times R)$$

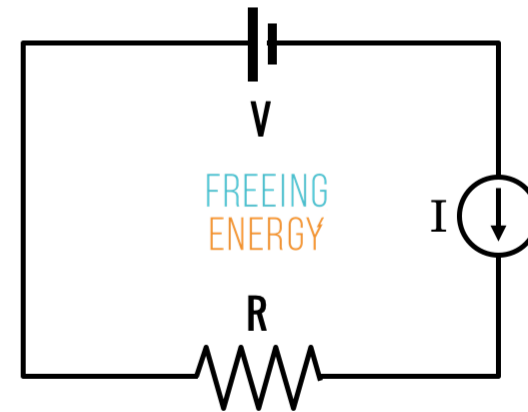
Water



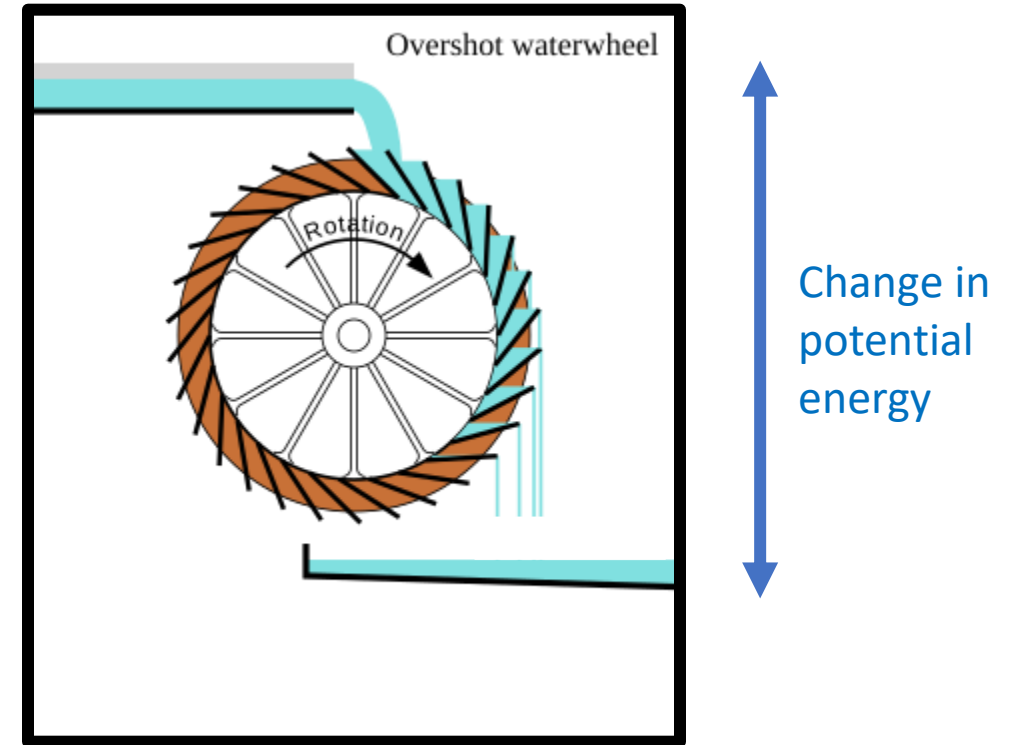
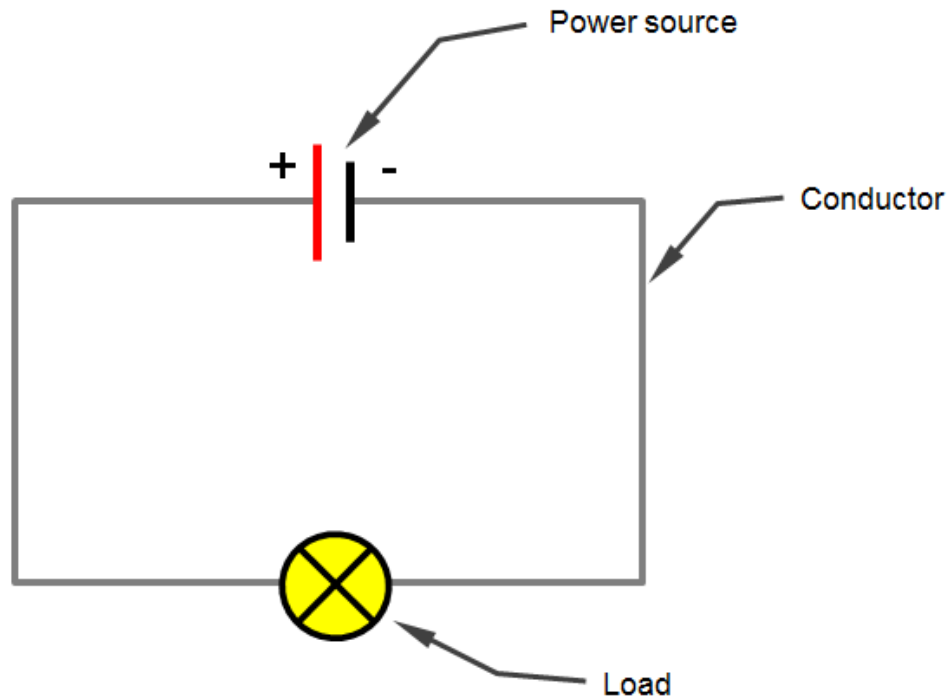
Electricity



Circuit Diagram



How does electricity drive a circuit element? A hydraulic analogy

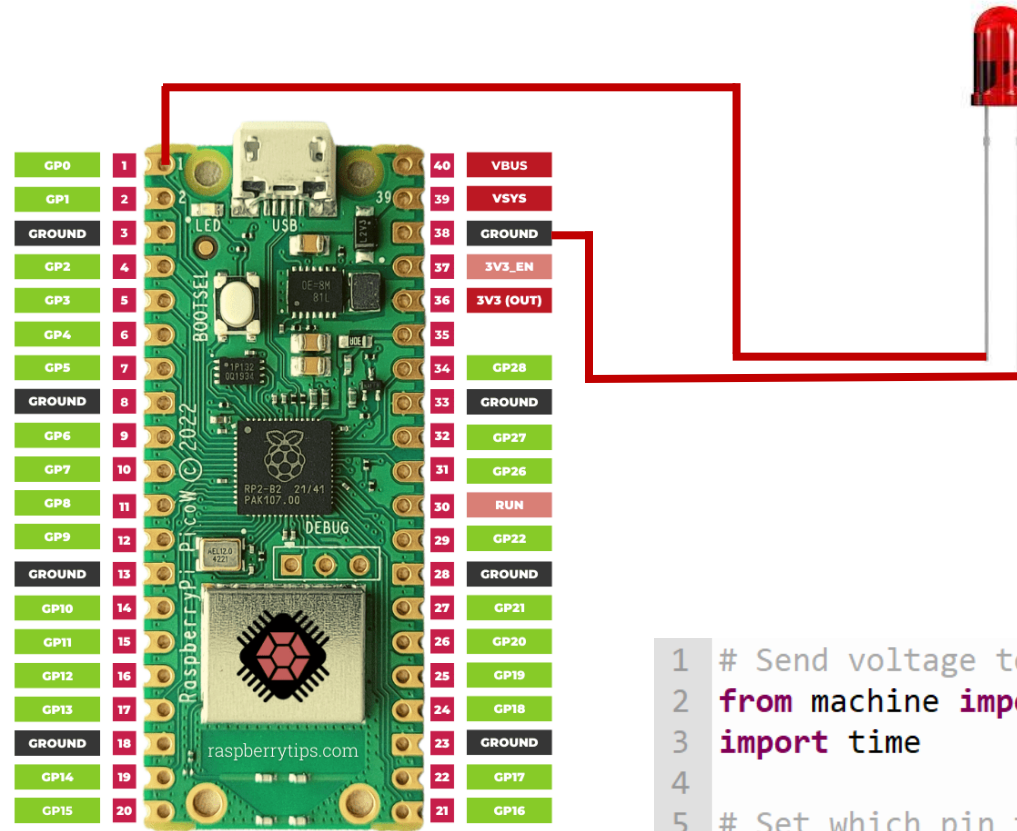
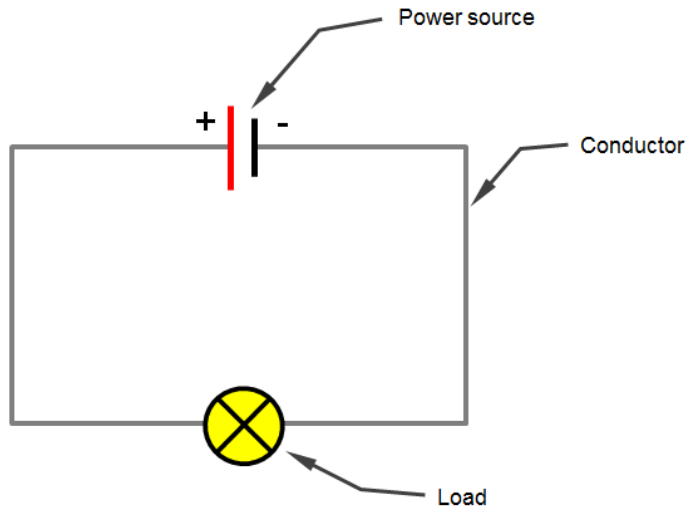


Change in potential energy drives a current

Potential energy: Voltage difference between two points on circuit
Current: flow of electrons through wire

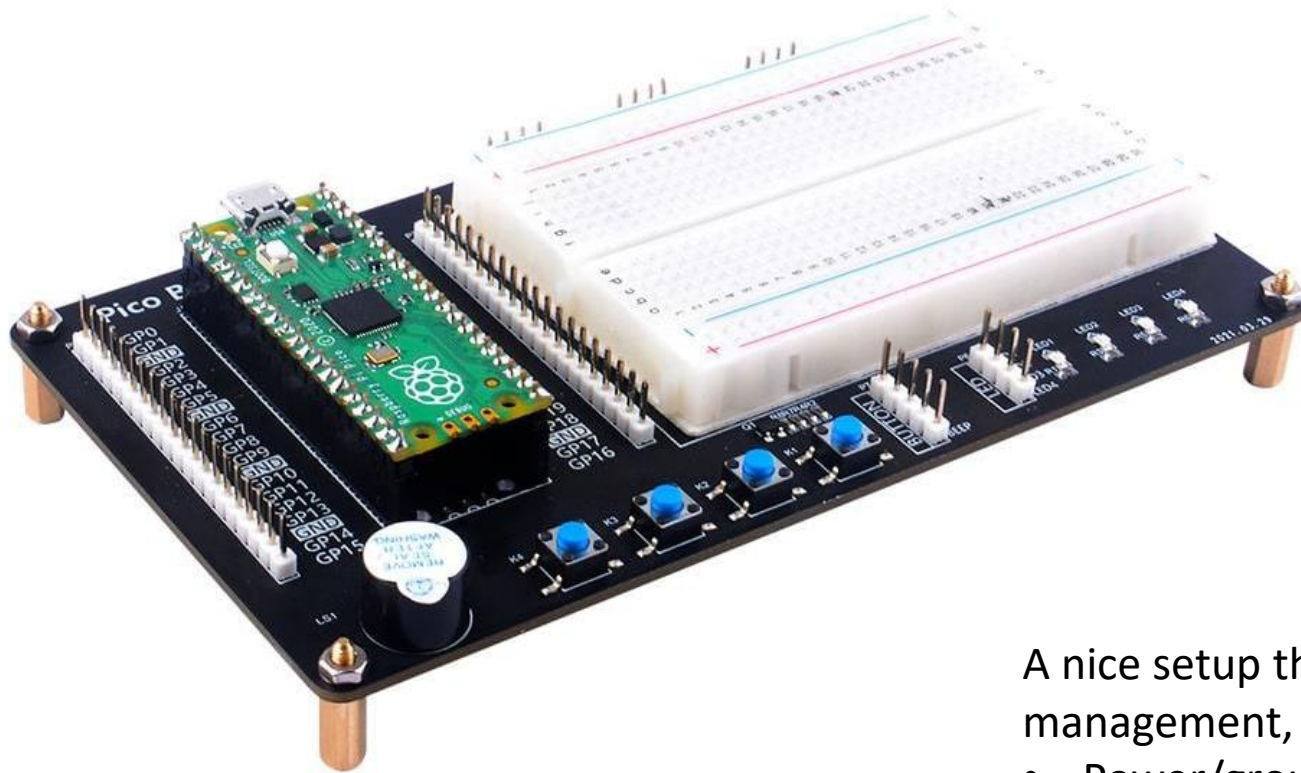
Potential energy: Difference in elevation storing gravitational energy
Current: flow of water through pipe

We will write a program that sets a voltage at one “pin” to switch lights off/on



```
1 # Send voltage to blink an LED
2 from machine import Pin
3 import time
4
5 # Set which pin to send voltage to
6 led = Pin('GP0', Pin.OUT)
7
8
9 while True:
10     led.toggle()      # set voltage off/on
11     time.sleep(0.5)   # wait a half second
```

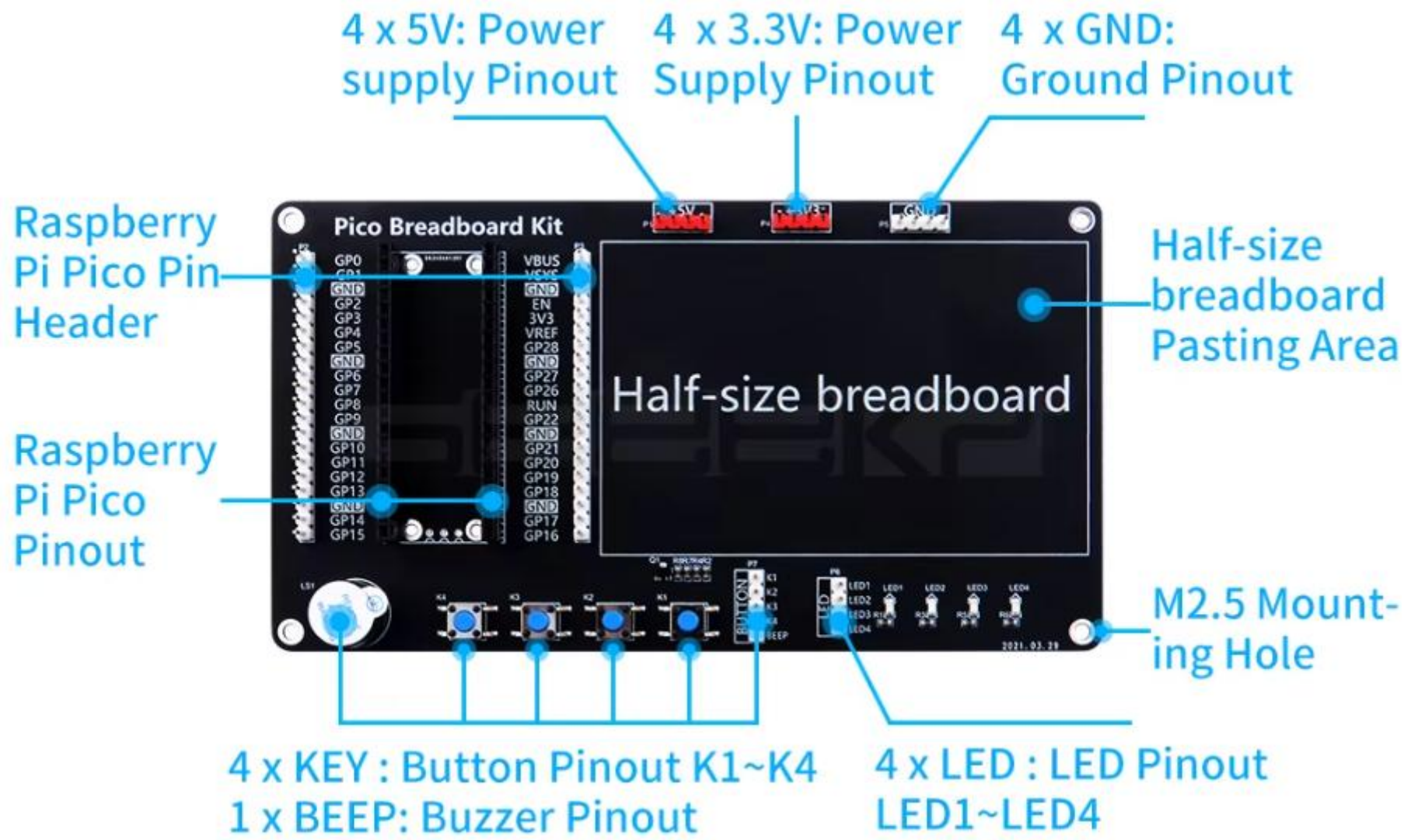
This is your Raspberry Pi Pico – a microcontroller with some basic functionality



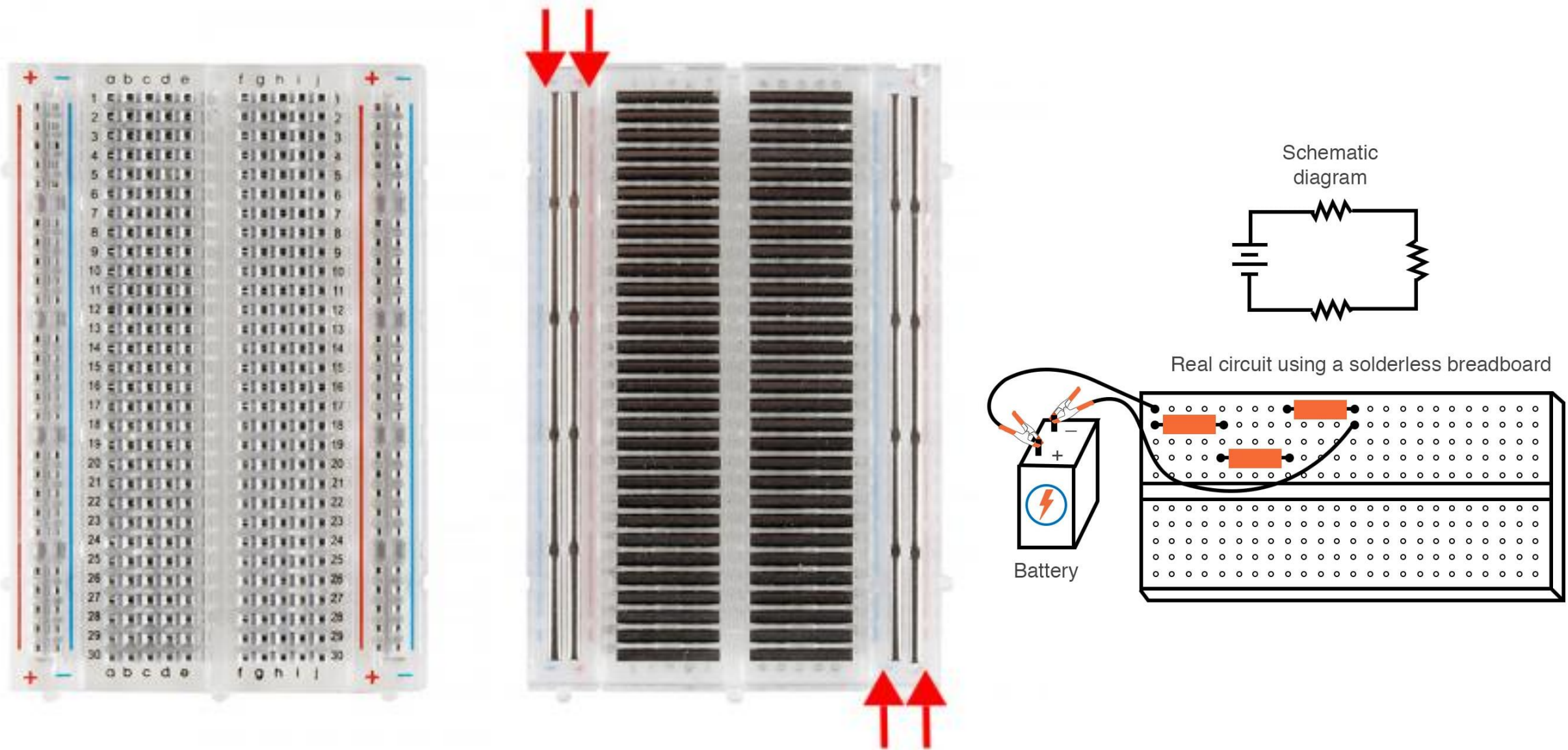
A nice setup that minimizes wire management, with built in:

- Power/ground lines
- Buttons
- LEDs
- Speakers
- Compact breadboard

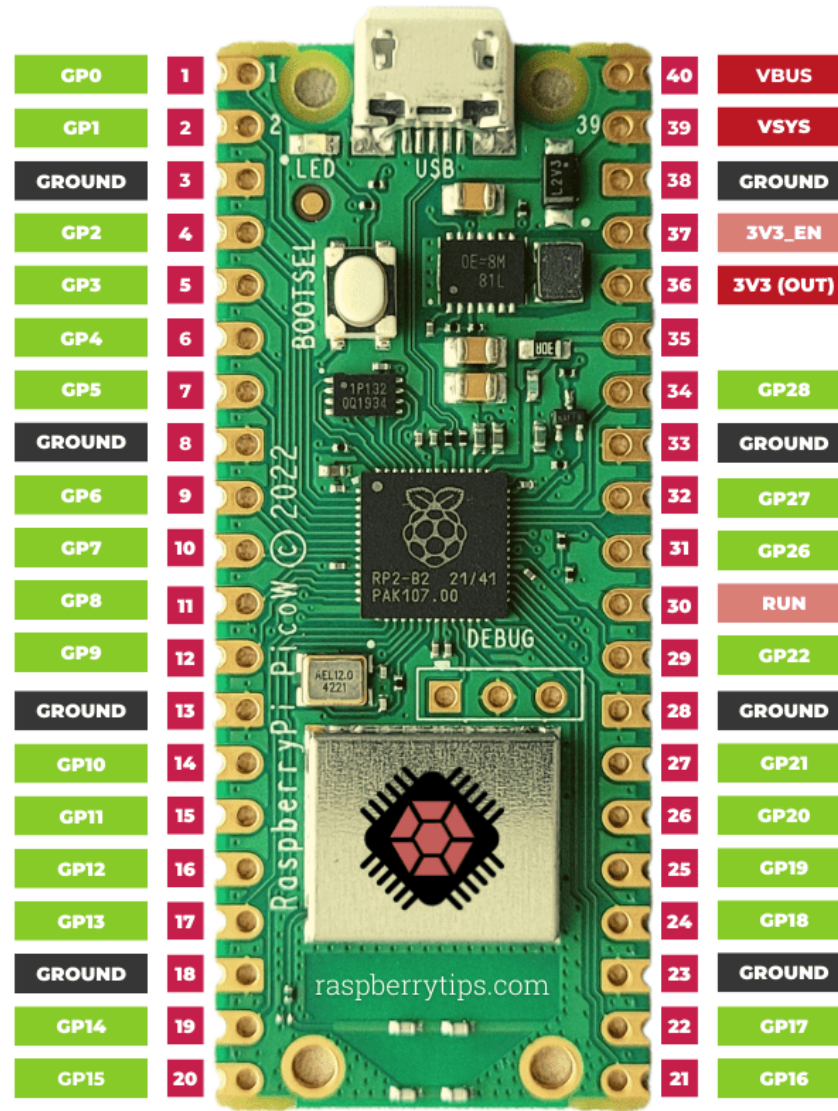
This is a mounting board that your Pico is attached to, with lights and convenient pins for connecting wires



This is a breadboard – it lets you build circuits without needing to solder wires



This is the actual Pico – we can use python to feed juice to pins



LED – built in LED on the pico can be lit without wiring for diagnostics

GPX – general purpose input output (GPIO), which the microcontroller can set to high or low voltage in output mode. Rated for 3.3V and 20mA.

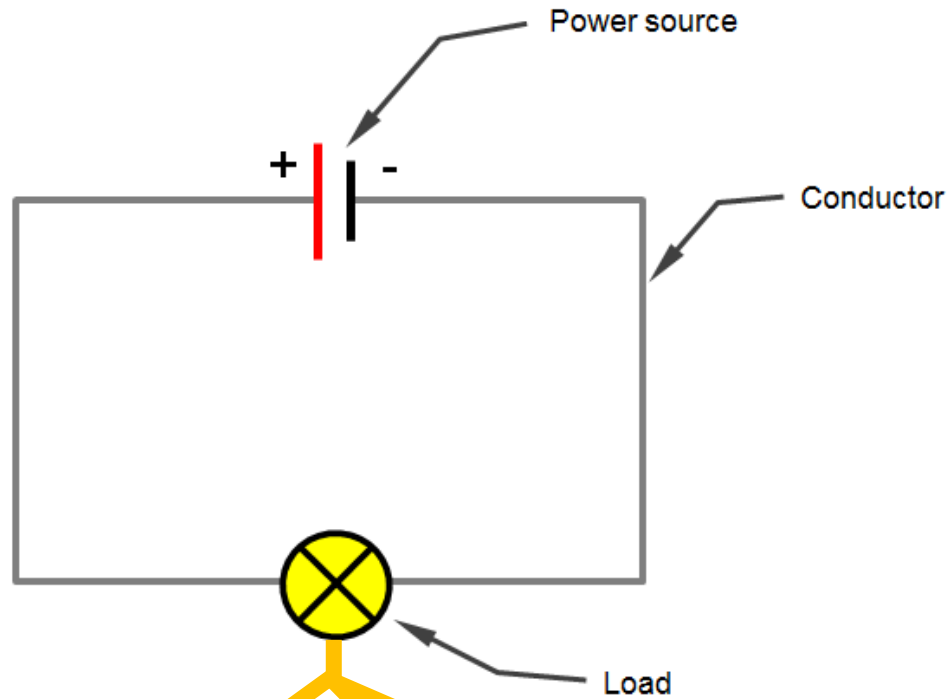
3V3 – A power source pin, which always outputs 3.3V and can draw up to 1A.

Ground – Where to close the circuit

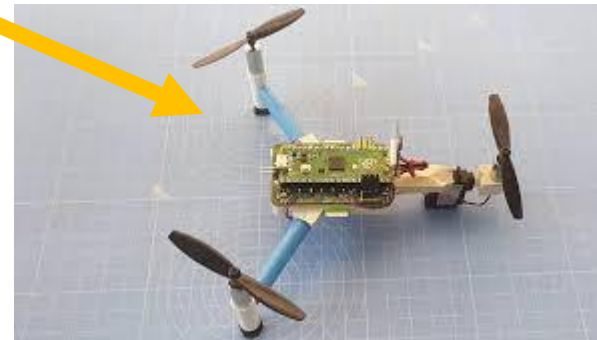
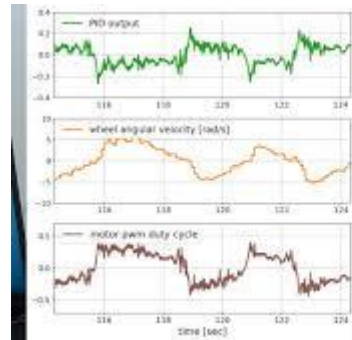
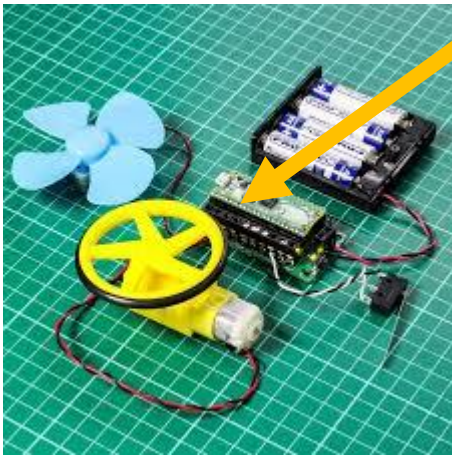
1. ADC pins – Only some pins can convert analog to digital. **Raspberry Pi Pico (RP2040):**

1. ADC0: GPIO26
2. ADC1: GPIO27
3. ADC2: GPIO28

Ready for labs



We now know enough python to write programs to drive loads and build some basic robots. Every couple weeks we'll break to play with robots in the GM lab.



Configuring your Pico for Thonny to see it

Install or update MicroPython (UF2)

Here you can install or update MicroPython for devices having an UF2 bootloader (this includes most boards meant for beginners).

1. Put your device into bootloader mode:
 - some devices have to be plugged in while holding the BOOTSEL button,
 - some require double-tapping the RESET button with proper rythm.
2. Wait for couple of seconds until the target volume appears.
3. Select desired variant and version.
4. Click 'Install' and wait for some seconds until done.
5. Close the dialog and start programming!

Target volume

family

MicroPython family

variant

version

info https://micropython.org/download/RPI_PICO2_W

Thonny - <untitled> @ 10:20

File Edit View Run Tools Help

<untitled> * x

```
1 from machine import Pin
2 import time
3
4 led = Pin(25, Pin.OUT)
5
6 while True:
7     led.value(1)      # 3.3 volts on the pin: LED on
8     time.sleep(0.5)   # wait half a second
9     led.value(0)      # 0 volts: LED off
10    time.sleep(0.5)
```

Shell x

```
Process ended with exit code 1.

Unable to connect to COM3: port not found

Process ended with exit code 1.
```

Local Python 3 • Thonny's Python

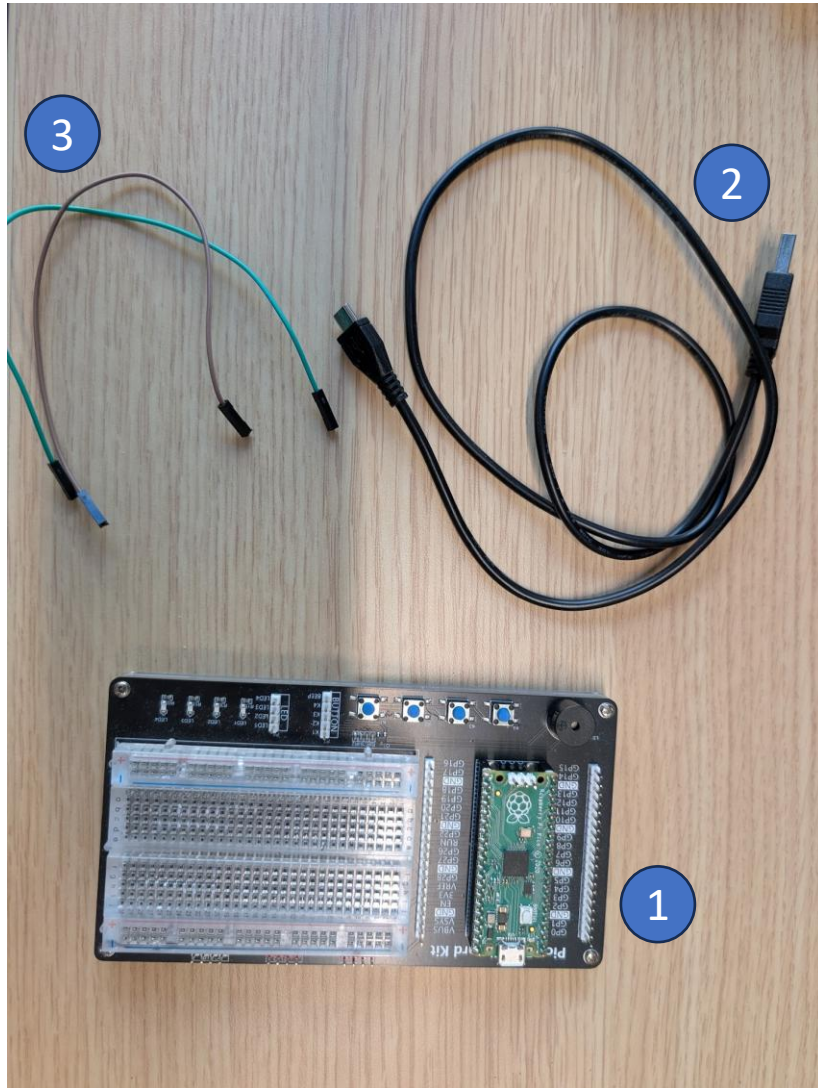
✓ MicroPython (Raspberry Pi Pico) • Board CDC @ COM3

MicroPython (RP2040) • Board CDC @ COM3

Configure interpreter...

MicroPython (Raspberry Pi Pico) • COM3

Material for today



1 Raspberry Pi Pico

2 USB Cable

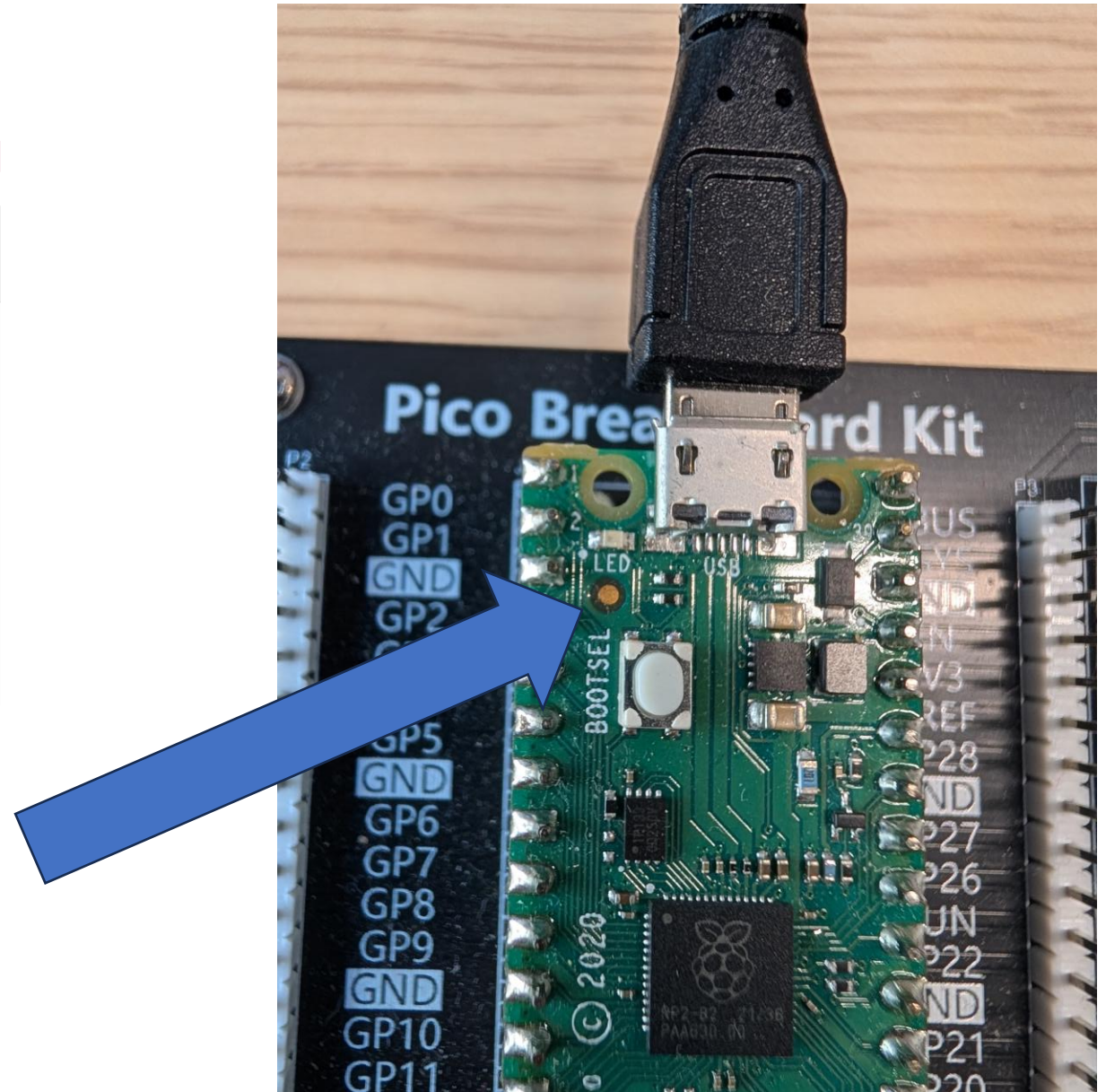
3 Female-to-female jumper cables

Connect the Pico to your computer

```
Thonny - <untitled> @ 10:20
File Edit View Run Tools Help

<untitled> * x
1 from machine import Pin
2 import time
3
4 led = Pin(25, Pin.OUT)
5
6 while True:
7     led.value(1)      # 3.3 volts on the pin: LED on
8     time.sleep(0.5)   # wait half a second
9     led.value(0)      # 0 volts: LED off
10    time.sleep(0.5)|
```

Running the above code will flash the built in LED on your board. This is how you can quickly confirm you are successfully executing code from your computer on the Pico.



Light up a real LED using a jumper

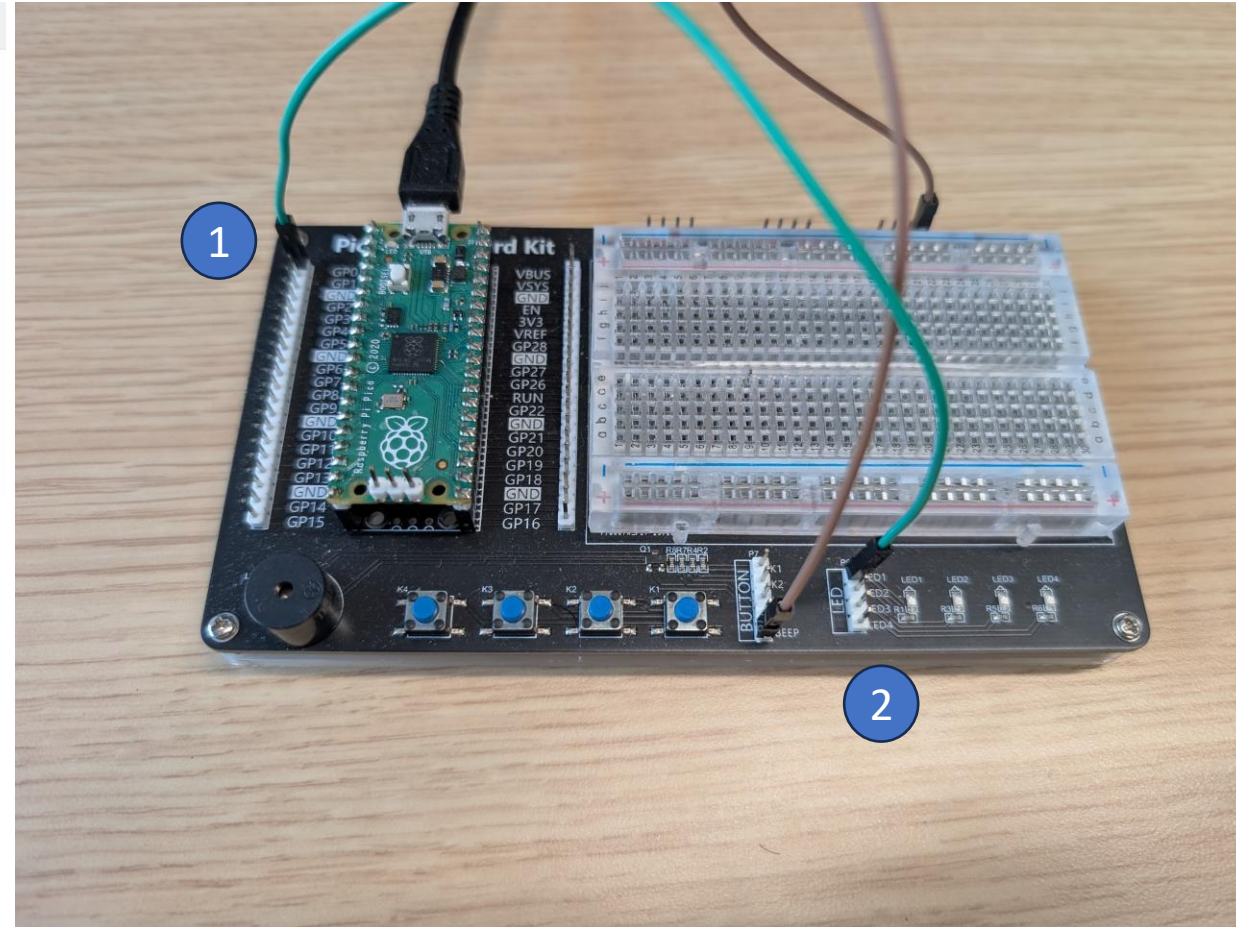
```
<untitled> * x
1 from machine import Pin
2 import time
3
4 led = Pin(0, Pin.OUT) #Switching the pin from 25 to 0 sends voltage to GP0
5
6 while True:
7     led.value(1)      # 3.3 volts on the pin: LED on
8     time.sleep(0.5)   # wait half a second
9     led.value(0)      # 0 volts: LED off
10    time.sleep(0.5)
```

1

Attach one end of the female-to-female jumper cable to GP0

2

Attach the other end to LED1 on the bottom right of your board



Today's canvas

- 1 Take a picture of your Thonny saying hello world
- 2 Take a picture of an LED lighting up

Extra challenge:

There are directions on the “Pico tutorial code” website to set up an actual LED on the breadboard. If you finish early, try to complete this to get some early practice learning how to wire circuits.

In class 5

✓ Published
Assign To
Edit
⋮

Today's class was held in the GM lab to get oriented with Thonny and the Picos.

For this assignment, take and upload two pictures on your cell phone of

- The Thonny outputting "Hello world" to your terminal.
- The LED light turned on on your Pico.

You can work in groups of two for this assignment to help each other get the Pico wiring set up, but everyone must have their own working Thonny on their own computers to be prepared for future lab sessions. You will submit individual homeworks for this exercise (i.e. not a group submission).

